What would make biodiversity offset markets work across all ecosystem types?

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Abstract

Biodiversity offset markets are meant to compensate, ecosystem by ecosystem, for habitats that development destroys. Do they? We analyse 1,124 transactions from Australia's largest mandatory offset scheme. Across 252 ecosystem types in the scheme, only 18% sustain a working market, and 47% of the threatened communities the scheme is designed to protect receive no direct investment at all. The reason is structural: buyers minimise cost, demand concentrates on a handful of cheap types, and developers route obligations through a government fund rather than purchasing credits directly. An agent-based model that reproduces this gap shows what would help. Restricting fund pay-in and requiring buyers to direct some demand toward underserved types raise coverage at modest cost; minimum prices and tighter government procurement rules do not. Markets that share this design risk the same failure unless procurement rules are rewritten.

Keywords: biodiversity offsets, ecological coverage, vegetation formations, competitive exclusion, procurement flexibility, biodiversity net gain

Biodiversity offset markets exist so that when development damages an ecosystem, the developer compensates by funding protection or restoration elsewhere. As of 2023, mandatory offset schemes transact an estimated $11.7 billion per year [1], with nineteen national and subnational schemes operational or in development across six continents [2]. Within each, a developer who damages a protected habitat or species must buy credits from a landholder who protects or restores an equivalent ecosystem elsewhere. Credits are specific to ecosystem type: a developer who clears woodland must buy woodland credits. The number and boundaries of these types are fixed by the regulator through official ecosystem classification lists, fragmenting each scheme

into hundreds of separate sub-markets. Because vegetation communities differ vastly in their spatial extent, landholder willingness, and restoration feasibility, the supply of credits varies dramatically across ecosystem types. The question is not whether these markets should be linked to development (that is their purpose) but whether they deliver compensation evenly across the ecosystem types they are mandated to protect. A 2019 global review of biodiversity offset schemes found that only approximately one-third achieved no-net-loss outcomes [3]. These failures have been attributed primarily to metric design, monitoring gaps, and political compromise [4, 5], but whether the market design itself systematically excludes most ecosystem types from receiving investment remains an open question.

There are theoretical reasons to suspect it does. Compliance buyers must offset impacts on a specific ecosystem type, but when their obligation can be met by credits from related types within the same vegetation formation, they minimise cost. Drechsler [6] predicted that this will concentrate demand in commercially accessible types, and zu Ermgassen et al. [7] argued that the resulting loss of scarcity signals undermines impact avoidance. But neither prediction has been tested with transaction-level data, and the role of developers routing demand through a government fund, rather than purchasing directly, has not been examined.

We test this using transaction-level data from the New South Wales (NSW) Biodiversity Offsets Scheme, Australia's largest mandatory offset market (AUD 105M annual turnover [8]), operational since 2017 and recording over 1,100 credit transactions across 252 ecosystem types. The scheme recognises 252 distinct ecosystem credit types, each corresponding to a specific vegetation community. Transaction data alone reveals that 112 of these 252 types (44%) have never been traded, but cannot explain why or predict which interventions would help. For that, we build an agent-based model in which four participant species interact across these credit types. Following Farmer's market-ecology framework [9, 10], species are defined by behavioural strategy rather than by institutional identity: (i) cost-minimising compliance demand: agents seeking the cheapest credit that satisfies their obligation; (ii) mean-reversion intermediation: agents bidding against rolling fundamental value to exploit price dispersion; (iii) thin-market procurement: budget-constrained agents that bid above the compliance median in low-liquidity types to ensure coverage; (iv) cost-floor supply: credit producers who sell only when market price exceeds a fraction of marginal production cost. In NSW these strategies are realised by, respectively, compliance buyers (i), intermediaries (ii), the Biodiversity Conservation Trust (BCT) (iii) drawing on the Biodiversity Conservation Fund (the Fund), and habitat banks (iv) (landholders who generate and sell credits). The model identifies three reinforcing mechanisms that explain why most ecosystem types are left without a functioning market, defined here as one with enough repeated transactions for buyers to reliably find a willing seller. First, compliance buyers select the cheapest available credit type, concentrating demand in a few well-supplied types (least-cost selection). Second, when buyers cannot find credits for their specific obligation, they pay into the Fund rather than purchasing directly, routing demand away from the types that need it most (fund routing). Third, credit types that have traded before are approximately 8.4 times more

likely to trade again, creating a self-reinforcing cycle in which popular types attract more demand while neglected types remain neglected (liquidity feedback).

Our model, which incorporates these three mechanisms alongside realistic agent behaviour and market structure, reproduces the observed pattern: most ecosystem types lack a functioning market. Because the model captures the mechanisms that generate the gap, we can use it to test which policy changes would close it. We evaluate five interventions, each targeting a different mechanism, and measure their effect and cost on *functional coverage*, the fraction of ecosystem types sustaining a functional market ($\geq$5 transactions). The most effective interventions are those that keep demand in the open market: restricting developers' access to the Fund (bypass reduction) and requiring a fraction of purchases to be directed toward underserved types (procurement flexibility). Unlike the buyer-choice flexibility that previous work has shown to erode scarcity signals [7], these are regulator-directed instruments that preserve price signals while channelling new demand where it is absent. Interventions that restrict flexibility or raise costs without redirecting demand, including setting minimum credit prices (price floors), requiring more credits for rare types (rarity multipliers), and mandating exact like-for-like matching for BCT purchases (BCT precision mandate), are ineffective or counterproductive. Because these policy conclusions could depend on how the model is built, we validate the results out of sample using a temporal split (calibrating on pre-2024 data and predicting post-2024 outcomes) and confirm them with an independent analytical model based on ecological competition theory.

The design failure documented here is not confined to NSW. At least nineteen national and subnational biodiversity offset schemes are operational or in development worldwide, including the United Kingdom's Biodiversity Net Gain framework, which became mandatory in 2024 [2]. Each operates a compliance market in which developers offset impacts on specific ecosystem types, and each faces the same structural risk: that least-cost compliance will concentrate demand before coverage gaps can be corrected. The NSW evidence shows that once the liquidity feedback cycle has set in, the pattern is self-reinforcing. The intervention window is open now, before these markets lock in the same design.

1 Results

Most ecosystem types lack a functioning market

Analysis of 1,124 priced transactions from the NSW Biodiversity Offsets Scheme public register (November 2019–March 2026) reveals a coverage gap. The Scheme recognises 252 distinct ecosystem credit types (offset trading groups, OTGs). Of these, 140 (56%) have recorded at least one priced transaction; the remaining 112 (44%) have never been traded. Only 45 (18%) have traded often enough to sustain what we term a *functional market*: five or more transactions over the observation period, a conservative threshold below which transaction frequency is too low for reliable price discovery [11] (Fig. 1a). We define *functional coverage* as the fraction of ecosystem types that clear this threshold. This gap is persistent, not transient: among ecosystem types below the functional threshold, transaction counts remain near zero across six years of data, and

no individual credit type shows a meaningful trend toward viability. The market is not self-correcting.

This coverage gap is not explained by the absence of potential supply: all 252 ecosystem types have registered landholders, confirming that credits could be generated if demand existed at viable prices. Nor is it explained by absent demand. Developers do incur offset obligations in rare credit categories, but instead of purchasing credits directly, they pay into the Fund, and the Trust purchases on their behalf. The 24 credit categories served only by the Trust have each received at least one such indirect purchase, consistent with developer demand existing for these types but being routed away from the open market via the Fund.

The exclusion is systematic, not random. Of the 252 ecosystem credit types, 79 correspond to listed Threatened Ecological Communities (TECs) and 173 to non-TEC vegetation types. Because offset obligations reference TECs by name, compliance demand flows disproportionately toward listed communities: TECs are significantly more likely than non-TECs to record at least one transaction, and under strict name-matching no non-TEC type appears in the priced-transfer register at all (Methods). But even this directed demand is insufficient. Of the 79 listed threatened communities, 47% receive no direct market investment: 31 have never been traded and 6 have been purchased only by the conservation Trust using Fund money, not by any compliance buyer on the open market. Only 18 sustain functional markets: the scheme fails to deliver reliable compensation for more than three out of every four threatened communities it lists.

The scale of fund routing is substantial. Of 1,097 credit retirements, 35% were Trust purchases on behalf of developers who paid into the Fund rather than purchasing credits directly [8]. Buyer concentration reinforces the pattern. The Herfindahl-Hirschman Index (HHI) reached 2,966 in 2024–25 (above the 2,500 threshold for highly concentrated markets), up from 2,187 the prior year [12] (Fig. 1b). The scheme does have a backstop for ecosystem types that the open market fails to serve, but as the next section shows, that backstop is itself under pressure.

The government backstop does not close the gap

The Biodiversity Conservation Trust (BCT, hereafter the Trust) is the NSW government agency responsible for investing in private land conservation. When developers cannot find credits in the open market for their specific ecosystem type, they can pay into the Fund and transfer the obligation to the Trust, which procures credits on their behalf. This makes the Trust the de facto safety net for thin-market ecosystem types: for 24 credit categories, it is the only buyer that has ever purchased credits. The discretion produces a measurable price premium: in thin markets (fewer than five lifetime transactions), the Trust pays a median of AUD 5,989 per credit versus AUD 1,125 for direct compliance buyers, a 5.3-fold differential that is highly significant (Methods). In functional markets the differential is markedly smaller, confirming that the premium reflects the Trust's role in thin markets rather than an artefact of overall price levels.

But this safety net is weakening. The Trust's share of credits more than doubled from 7% in 2022–23 [13] to 16% in 2024–25 [12], yet the fraction of its purchases matching the specific credit type required by the underlying developer obligation

(the like-for-like rate) fell from 90% to 65% over the same period. If volume growth comes at the cost of reduced like-for-like precision, the Trust may be processing Fund-routed demand rather than directing procurement toward conservation targets, and the 24 rare credit categories that depend on exact matching go unserved. A developer who damages Coastal Upland Swamp pays into the Fund expecting compensation for that specific community; the Trust may instead procure credits from a more readily available type, leaving the original ecosystem uncompensated.

The transaction data reveal what is happening but not why, nor which interventions could improve outcomes. We test five: restricting developers' access to the government fund, redirecting a fraction of demand toward underserved types, requiring higher credit quantities for rare ecosystem types, setting a minimum credit price, and mandating exact matching for the government procurement agency. Together these target each component of the coverage gap from both the demand and supply sides.

Keeping demand in the open market is the most effective intervention

To test whether the observed coverage gap arises from market structure rather than idiosyncratic factors, we built an agent-based model of the NSW scheme: 252 ecosystem credit types grouped by 15 vegetation formations across 40 ecological subregions, with four kinds of participant: compliance buyers (cost-minimising), intermediaries (profit-seeking), the Trust (procurement-flexible), and habitat banks (credit suppliers). The model incorporates a self-reinforcing demand-concentration mechanism in which credit types that have traded before attract more demand, with weights estimated directly from the NSW transaction register: types with $\geq$10 prior transactions are 8.4$\times$ more likely to be chosen than untraded types (Methods). All other parameters are derived from data; one parameter (the rate at which compliance buyers attempt within-formation substitution) is calibrated to match the observed Fund routing rate.

The model reproduces the empirical coverage gap. Functional coverage in the model is 16.3% (median across 50 Monte-Carlo runs; observed: 17.9%), the simulated and observed shares of transactions across vegetation formations are tightly correlated (Methods, Fig. 2a), and the simulated split of retirements between compliance buyers and the Trust (65.8%/34.2%) reproduces the observed 65%/35% breakdown. Five formations that are empirically near-extinct in the data (Grasslands, Rainforests, Saline Wetlands, Arid Shrublands, and Heathlands; each $\leq$9 lifetime transactions) emerge as near-extinct in the model without being targeted by calibration.

We test five policy interventions, each targeting a different component of the coverage gap (Table 1; Fig. 2b). Only two of the five interventions improve coverage relative to baseline (16.3%), and both work by keeping demand in the open market rather than diverting it to the Fund. Bypass reduction (making it harder for developers to pay into the Fund) raises functional coverage to 17.5% and cuts Fund routing from 34.1% to 21.3%. Procurement flexibility mandates (requiring compliance buyers to direct 20% of purchases toward thin-market types) raise coverage to 17.1%. Three interventions fail. Price floors (AUD 3,000 minimum) push more developers into the Fund (routing rises to 50.5%) without generating new direct demand. Mandating exact like-for-like matching for the Trust's purchases is actively counterproductive (coverage falls to

12.7%): the Trust can no longer procure credits when the exact ecosystem type has insufficient supply, and damage falls disproportionately on threatened communities (in the model, FC-TEC drops from 25.3% to 16.5%, eliminating a third of the functional TEC markets in simulation; empirical FC-TEC is 22.8%). The rarity multiplier (2× credits required for rare types) has only a marginal effect (16.5%), confirming that the bottleneck for rare types is the existence of demand, not the quantity per transaction. One formation (Arid Shrublands, Acacia sub-formation) remains market-extinct under all scenarios: its development pressure is so low that no demand-side intervention generates obligations there.

Robustness

The conclusions are robust to specification, sample, and modelling framework [14]. Calibrated to a single observable (the 35.4% Fund routing rate), the model reproduces three other empirical patterns it was not calibrated to: the headline coverage gap, the compliance/Trust split of retirements, and the dominance of Grassy Woodlands. The formation-level ranking is preserved under both a within-sample temporal split (pre-2024 vs 2024–26) and a formation hold-out (excluding Grassy Woodlands from calibration). Parameter sensitivity confirms that functional coverage moves by less than one percentage point when any parameter is perturbed across a six-fold range; the only exception (the number of Trust agents, 5.4 pp) does not change the scenario *ranking*. A separate analytical model based on Generalised Lotka–Volterra ecological dynamics, calibrated to the same data but solved as differential equations rather than agent simulation, identifies the same most- and least-effective interventions (Supplementary Note 1; Supplementary Fig. 1a–b; Supplementary Table 6).

2 Discussion

Intervention hierarchy. The five policy scenarios test interventions targeting each component of the coverage gap. Two clear lessons emerge. First, the most effective interventions are those that keep demand within the open market rather than diverting it to the Fund. Bypass reduction (functional coverage 17.5%) and procurement flexibility (17.1%) both work by increasing the probability that a compliance buyer transacts directly with a credit supplier, rather than paying into a government fund. This finding converges with the NSW regulator's own assessment: IPART [13] independently concluded that the Fund pay-in option prevents the market from developing and recommended phasing it out. Our model quantifies why: every obligation that enters the Fund is one fewer direct transaction in the open market, and the cumulative effect is a 13 percentage point reduction in Fund routing (from 34.1% to 21.3%) when bypass is restricted.

Bypass reduction operates as a necessary precondition rather than a stand-alone fix. The 2024 Amendment Act acquittal deadline reduced NSW market bypass from 81% in 2022–23 to approximately 18% in 2024–25 [12], while functional coverage held at 18% over the same window (Supplementary Table 1 footnote): bypass reduction

redirects demand toward the open market, and procurement flexibility — which allocates a share of compliance demand to underserved types — translates that redirected flow into coverage gains.

The US 2008 Final Rule [15] reached a similar conclusion for wetland mitigation, establishing a preference hierarchy (mitigation banking > in-lieu fee) after a decade of evidence that fund-based mechanisms delivered weaker ecological outcomes [16]. The closest precedent for procurement flexibility mandates is technology carve-outs in Renewable Portfolio Standards, where mandating minimum purchases from specific energy types increased technology diversity even when general portfolio targets did not [17].

Second, interventions that restrict flexibility or raise costs without redirecting demand are ineffective or counterproductive. Price floors cause developers to route more obligations into the Fund (routing rate from 34.1% to 50.5%) without generating new direct demand [18]. Mandating exact like-for-like matching for BCT purchases reduces coverage to 12.7%, confirming that government intermediaries need procurement flexibility to serve thin markets [19]. The rarity multiplier (2× credits for rare OTGs) has negligible effect (16.5%) because the bottleneck for underserved types is the existence of demand, not the quantity per transaction; this is consistent with the theoretical finding that uniform multipliers are inadequate for rare habitats [20].

zu Ermgassen et al. [7] argued that relaxing like-for-like rules destroys the scarcity signal that incentivises impact avoidance, and Drechsler [6] showed formally that buyer-choice flexibility always reduces ecological benefit. Our data provide empirical evidence consistent with this prediction. The procurement flexibility mandate does not relax like-for-like rules for the buyer; instead, regulators direct a fraction of compliance demand toward underserved types, preserving the scarcity signal for the remaining 80% of purchases while channelling new demand where it is absent. Bypass reduction and procurement flexibility therefore constitute the core policy package for improving ecological coverage. Both can be implemented within existing statutory frameworks.

Why these interventions work: three layers of exclusion. The effectiveness of demand-side interventions reflects the depth of the coverage gap, which operates at three levels. First, non-TEC vegetation types (the ecological matrix that provides connectivity and buffering for threatened patches [21, 22]) receive systematically less direct market investment than listed TECs, because offset obligations reference TECs by name (Methods). Second, even when an obligation does fall on a TEC, current rules let the offset be discharged outside the open market: offset obligations capture only direct clearing impacts, while supply-chain, cumulative, and diffuse impacts remain uncompensated [23]. Third, and most critically, the market fails even within its own intended scope: 47% of listed TECs receive no direct market investment (37 of 79 have never traded or are served only by the Trust), and the 24 ecosystem types served only by the Trust show that developers route demand through the Fund rather than the open market. The market is failing on its own terms [24]: designed to compensate for development impacts on listed threatened communities, it instead channels investment toward the most commercially accessible types. zu Ermgassen et al. [7] noted that the 2017 Biodiversity Conservation Act places no legal restriction on the Trust offsetting using any habitat type anywhere in the state; our data quantify the consequence. The

Trust's budget (AUD 47.3M per year) is sufficient to sustain market activity in all but one formation, yet five are empirically near-extinct, consistent with the Trust's declining like-for-like precision (90%→65%).

Costs and implementation. Cost-effectiveness — private outlay per percentage point of functional coverage gained — is modest under the effective interventions. Bypass reduction increases total market expenditure by 3.1% and procurement flexibility by 0.6% (Supplementary Table 8), with average credit prices essentially unchanged. The Trust's procurement load follows the Fund routing rate: bypass reduction shifts routing from 34.1% to 21.3%, reducing regulator outlay; price floors increase routing to 50.5% and so raise the regulator's burden even where private cost is flat. By comparison, a 20% solar carve-out in US Renewable Portfolio Standards increased programme costs by 28% [25]. A striking asymmetry emerges: it is far easier to damage coverage than to improve it. The BCT precision mandate reduces coverage by 3.6 percentage points (eliminating a third of functional TEC markets), while the best positive intervention (bypass reduction) gains only 1.2 percentage points. This asymmetry implies that poorly designed procurement rules can cause irreversible coverage losses that even well-designed interventions struggle to reverse. Flexibility mandates also raise acquittal complexity for developers, and the 20% threshold invites gaming near the margin. An alternative route to the same goal is to embed coverage directly into the credit metric itself, for example through irreplaceability-weighted credits that assign higher value to underrepresented habitat types [26]. Functional coverage should become an explicit regulatory metric [27], monitored alongside volume and price, to provide an operational early warning of coverage decline before it becomes irreversible.

Generalisability and limitations. The demand-concentration mechanism is not unique to biodiversity markets: analogous coverage biases have been documented in renewable energy markets [28, 29] and water entitlement markets [30, 31], where mandatory or compliance-driven demand also concentrates on the cheapest types. The mechanism here is therefore structurally general, but the quantitative findings are NSW-specific: every parameter is calibrated to NSW data, and the four-participant typology presupposes a compliance-buyer population, an exogenous-mandate demand structure, a government safety-net agency, and credit-type segmentation. These design features describe most current compliance offset schemes, including the United Kingdom's Biodiversity Net Gain framework, but transferring quantitative numbers to markedly different designs would require recalibration.

The scenario ranking (Bypass Reduction > Procurement Flexibility > Rarity Multiplier ≥ Baseline > Price Floor > BCT Precision Mandate) holds across the parameter ranges examined in Supplementary Table 6 and the global sensitivity analysis in Supplementary Note 2.

Functional coverage in this analysis is an outcome metric tied to the regulator's stated intent — every listed ecosystem type should sustain a market — rather than a welfare measure integrating ecological benefit against fiscal cost; the ranking reports how well each intervention satisfies the scheme's compensatory design. Scenario-comparison percentiles overlap by approximately one percentage point, so the ranking is ordinal in the small-difference range. The next test of generalisation comes from the United Kingdom's Biodiversity Net Gain framework, which became

mandatory in 2024. Making biodiversity offset markets work across all ecosystem types requires procurement rules that keep demand in the open market rather than rewarding cost minimisation. For NSW and every market that follows it, the intervention window remains open, the policy tools are identified, and the statutory authority to act already exists.

Online Methods

Market ecology framework. The market ecology framework treats financial market participants as biological species competing for limited profit opportunities, with population dynamics governed by the profitability of their behavioural strategies [9, 10]. The framework extends the ecological metaphor of Lo's Adaptive Markets Hypothesis [32] into a formal dynamical system, drawing on earlier applications of ecological network theory to financial stability [33, 34]. We build on the formulation of Scholl, Calinescu, and Farmer [10], who showed that the ecological community matrix of interacting trading strategies can be computed numerically from a Walrasian agent-based model. We use Generalised Lotka-Volterra (GLV) equations as an analytically tractable complement to the ABM, validated by computing the community matrix from ABM perturbations following their methodology (Supplementary Note 1; Spearman $\rho = 0.69$, $p < 10^{-30}$). Each participant type i is represented as a biological species with population $N_i(t)$ (market capital, AUD), intrinsic growth rate r_i, carrying capacity K_i, and pairwise interaction coefficients α_{ij} encoding how species j affects the per-capita growth of species i. The mean-field GLV dynamics [35] are:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{\sum_j \alpha_{ij} N_j}{K_i}\right) + \sigma_i N_i \xi_i(t), \tag{1}$$

where $\xi_i(t)$ is independent standard Brownian noise with intensity σ_i, integrated via the Euler-Maruyama method ($dt = 0.1$ months). The competitive exclusion principle [36] provides the key analytical prediction: when two species share a sufficiently similar niche (α_{ij} large), the inferior competitor will be excluded in the long run. The GLV system is used for analytical scenario assessment and sensitivity analysis; primary results use the calibrated agent-based model described below.

Species classification and micro-founded demand. Four functional participant species are defined following the principle that species are grouped by behavioural strategy, not institutional identity [9**?** , 10] (Supplementary Table 3). Each species is identified by a signal function $\phi_i(\text{state})$ that maps observable market state to the agent's demand: cost-minimising compliance, $\phi = -\text{price}$ within feasible within-formation OTGs; mean-reversion intermediation, $\phi = (p_{\text{fv}} - p)/(\gamma\sigma^2)$; thin-market procurement, $\phi = \mathbb{1}[\text{liquidity} < L^*] \cdot (\bar{p}_{\text{thin}} - p)$; cost-floor supply, $\phi = \mathbb{1}[p > 0.8c]$ where c is marginal production cost. Two distinct organisations are the same species if and only if they share ϕ_i. Demand functions are grounded in constant absolute risk aversion (CARA) utility maximisation [37] with heterogeneous beliefs, following Brock and Hommes [38] and Chiarella [39]; the equivalence of the CARA derivation and the bid rules below is established in Supplementary Note 1.2 (Spearman $\rho \geq 0.97$ across 50 MC seeds):

1. **Compliance buyers.** Developers with statutory offset obligations. Demand follows a logistic willingness-to-pay schedule: $z(p) = \theta \cdot \text{remaining} \cdot [1 + e^{\gamma(p-\bar{p})}]^{-1}$, where $\bar{p}$ is the agent's reservation price (lognormal, $\mu_{\log} = 9.0$, $\sigma = 0.4$; AUD) and $\theta \sim U(0.8, 2.0)$ is individual aggression.
2. **Intermediaries.** Entities appearing on both sides of transactions; encompasses brokers and price-volatility-exploiting intermediaries. CARA demand [37, 39]: $z(p) = \theta \cdot (p_{\text{fv}} - p)/(\gamma\sigma^2)$, clipped by a leverage constraint $|z \cdot p| \leq 2W$, where p_{fv} is a rolling mean of recent clearing prices.
3. **BCT / procurement-flexible buyers.** Budget-constrained agents with fixed monthly capital allocations (lognormal, $\mu_{\log} = \log 326{,}389$, $\sigma = 0.5$; AUD), derived from BCT's 45% share of total market value [12] divided by 12 agents × 12 months. BCT is modelled as risk-neutral following Arrow and Lind [40], with procurement uncertainty captured through the empirically calibrated bid ceiling (AUD 5,989). The 11–17% risk premium charged by the Biodiversity Conservation Fund reflects administrative and procurement uncertainty costs, not utility-theoretic risk aversion. Unlike compliance agents, BCT-type agents do not select the cheapest available OTG; they are willing to pay above the compliance median for credits in thin markets.
4. **Habitat banks.** Supply-side agents holding initial credit inventories and producing new credits at marginal costs drawn from lognormal distributions ($\mu_{\log} = 7.62$, $\sigma = 0.5$, corresponding to the first quartile of observed ecosystem credit prices, AUD 2,033). Sell when market price exceeds 80% of production cost, consistent with cost-driven supply in thin environmental markets [41].

Population structure. Strategy populations are determined by the regulatory structure of the scheme: 30 compliance buyers, 12 BCT-type agents, 10 intermediaries, and 30 habitat banks per simulation. Compliance demand is regulatorily mandated and BCT operates under a public mandate, so the model captures within-formation and across-formation competition for finite compliance demand directly — the dynamic the GLV reduction characterises — rather than between-strategy capital flows.

Formation-level interaction matrix. The 15 × 15 interaction matrix $\boldsymbol{\alpha}$ (equation 1) encodes how vegetation formations compete for a shared pool of compliance demand. Interaction strength α_{ij} between formations i and j is derived from two empirical quantities: geographic overlap (Jaccard index of IBRA subregion co-occurrence) and price competition (cheaper formations suppress demand for expensive ones more strongly). Diagonal elements $\alpha_{ii} = 1$ (intraspecific self-limitation). The strongest off-diagonal interactions occur between Dry Sclerophyll Forests and Grassy Woodlands, which share the most subregions and compete for the same compliance demand (Supplementary Fig. 1a).

Formation-level calibrated ABM. The primary model operates on the 252 real ecosystem OTGs grouped by 15 vegetation formations and 40 IBRA subregions. OTG-level data (transaction counts, median prices, registered credit supply, official TEC status) are loaded from the NSW public registers. Offset obligations are generated by first drawing an IBRA subregion weighted by observed development pressure (transaction volume per subregion), then a formation within that subregion, then a specific TEC OTG using power-law concentration weights. Non-TEC OTGs

receive zero direct compliance obligations, reflecting the regulatory structure in which planning instruments reference listed TECs.

Four agent types interact monthly: 30 compliance agents (least-cost selection among OTGs within their obligation formation), 10 intermediaries (concentrate in liquid formations), 12 BCT-type agents (procurement-flexible, can purchase across formations with a preference for thin-market OTGs), and 30 habitat banks (invest in credit production where demand signals exist). The model runs for $T = 72$ months (matching the observation window). Results are computed across 50 Monte Carlo seeds.

Liquidity feedback mechanism. Each OTG tracks its cumulative transaction count. The probability that a compliance or intermediary agent considers a given OTG is weighted by empirical transition probabilities computed from the NSW transaction register: OTGs with zero prior transactions receive a base weight of 1.0; those with 1–2 prior transactions receive 1.5×; 3–4 receive 2.9×; 5–9 receive 5.1×; and 10+ receive 8.4×. BCT agents are exempt from this feedback, reflecting their institutional mandate to reach thin-market OTGs. These weights are empirically estimated transition probabilities from the NSW transaction register and are treated as a fixed preferential-attachment kernel.

Data-derived parameters. All model parameters are derived from the NSW transaction and supply registers or calibrated to a single observable datum (the Fund routing rate). Parameter derivations are documented in Supplementary Table 5 and computed by `scripts/derive_parameters.py`. No ad-hoc values are used. Sensitivity analysis (Supplementary Table 6) confirms that functional coverage varies by less than 1 percentage point when any parameter is perturbed across a 6-fold range, except for the number of BCT agents (5.4 pp), which is constrained by institutional design. Importantly, the scenario ranking (Bypass > Procurement Flex > Rarity ≥ Baseline > Price Floor > BCT Precision) is preserved across the full n_{BCT} sweep, even though FC levels move 5.4 pp; results that depend on rank order are therefore robust to BCT-agent specification.

Calibration and validation. The NSW BOS public transaction register was accessed from the NSW Government Land Management and Biodiversity Conservation portal (accessed March 2026). After removing transactions with missing prices and those below AUD 100, 1,124 priced transactions from November 2019–March 2026 remain: ecosystem credits $n = 764$ (68%); species credits $n = 360$ (32%). OTG names in the transaction register do not exactly match names in the supply register due to encoding differences; buyer-type classification (BCT vs compliance) uses a supply-register matching step that resolves 4 borderline cases (documented in *docs/DATA_CONTRACT.md*). Buyer concentration statistics and annual market volumes are drawn from IPART Discussion Papers [8] and the NSW Audit Office [42].

Compliance agents follow partial like-for-like rules: when their obligation OTG has no supply, they attempt a within-formation substitution with probability $P_{\mathrm{variation}} = 0.5$; otherwise the obligation is routed to the Biodiversity Conservation Fund, where BCT agents fulfil it. The variation parameter is the sole calibration target, selected to match the observed Fund routing rate of 35.4% (model: 34.1%; Supplementary

Table 5). $P_{\text{variation}}$ is also the lever that bypass reduction perturbs: the bypass-reduction scenario raises it from 0.5 to 0.8. The +1.2 percentage-point gain in functional coverage under bypass reduction should therefore be read as the response of a single calibrated parameter within its sensitivity range (FC = 15.5–17.5% as $P_{\text{variation}}$ varies in $[0.3, 0.7]$; Supplementary Table 6), rather than as an effect identified independently of the calibration. Calibration validation: (i) functional coverage among TECs (25.3%, observed: 22.8%) and non-TECs (12.1%) emerge without being calibration targets; (ii) the formation-level transaction ranking matches observations (Spearman $\rho = 0.98$, $p < 10^{-10}$); (iii) the compliance/BCT obligation split (65.8%/34.2%) reproduces the observed 65%/35% retirement breakdown; (iv) model functional coverage = 16.3% (10th–90th: 14.6–17.9%) is within 1.6 percentage points of the observed 17.9%. Full data provenance is documented in *docs/DATA_SOURCES.md*.

Policy scenarios. Five interventions modify procurement rules relative to the calibrated baseline: (1) *Bypass reduction*: within-formation substitution probability raised from 0.5 to 0.8, making it harder for developers to route obligations through the Fund; (2) *Procurement flexibility*: 20% of compliance obligations redirected to OTGs below the functional threshold, regardless of formation; (3) *Rarity multiplier*: developers must retire 2× credits for OTGs below the functional threshold (<5 cumulative transactions); (4) *Price floor*: minimum price AUD 3,000 for ecosystem credits in sub-threshold OTGs; (5) *BCT precision mandate*: BCT agents restricted to exact like-for-like purchases only (no cross-type substitution). The primary outcome metric is *functional coverage*, the fraction of OTGs that record five or more transactions across the simulation horizon, disaggregated by vegetation formation and TEC status. Scenario analysis uses 50 Monte Carlo seeds per scenario.

Robustness. Qualitative conclusions are validated following Fagiolo et al. [14]: (i) pattern reproduction (four non-targeted empirical patterns); (ii) within-sample temporal split (pre-2024 calibration, post-2024 prediction); (iii) global Sobol sensitivity analysis across the policy-relevant parameter space (Supplementary Note 2); (iv) cross-model agreement with a GLV mean-field model with 15 formations as competing populations (equation 1; Supplementary Fig. 1a–b; Supplementary Table 7). The like-for-like variation parameter ($P_{\text{variation}}$) is calibrated to a single observable datum (the 35.4% Fund routing rate); the other qualitative conclusions emerge without targeted calibration (Supplementary Table 6).

Code availability. All model code is available at https://github.com/jl-resendiz/biocredit. The production formation-level ABM (50 Monte Carlo seeds, 72 timesteps, all five policy scenarios and the baseline) is implemented in `scripts/run_abm.py`; the GLV analytical complement in `src/model/glv_formation.py`; shared infrastructure (entity classes, configuration) lives under `src/model/`. Pinned dependency versions are in `requirements.txt`; floors are Python 3.10+, NumPy ≥1.24, SciPy ≥1.10, Pandas ≥2.0.

Use of AI tools. During the preparation of this manuscript, the author used Claude (Anthropic) for code review and manuscript formatting. The author reviewed and edited all outputs and takes full responsibility for the content of the publication.

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Declarations

- **Funding:** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Competing interests:** The author declares no competing interests.
- **Ethics approval:** Not applicable (no human or animal subjects).
- **Data availability:** NSW BOS transaction register data are publicly available at the NSW Government Land Management and Biodiversity Conservation portal. All processed data used to produce the figures and tables are provided as Supplementary Data.
- **Code availability:** All model code, including the formation-level ABM, GLV engine, and analysis scripts, are available at https://github.com/jl-resendiz/biocredit under an MIT licence.
- **Author contributions:** J.L.R. conceived the study, developed all models, performed calibration and analyses, and drafted the manuscript.

Table 1 Policy scenario outcomes from the calibrated formation-level ABM (50 Monte Carlo seeds; 10th–90th percentile in brackets). Functional coverage (FC) = fraction of 252 OTGs sustaining ≥5 transactions. Fund rate = fraction of obligations routed through the Biodiversity Conservation Fund (observed: 35.4%). Cost Δ = change in total market expenditure relative to Baseline. Bypass reduction and procurement flexibility are the two most effective interventions; price floors and BCT precision mandates are counterproductive.

Scenario	FC (ABM)	Fund rate	Cost Δ	Mechanism targeted
Bypass Reduction	17.5% [15.5–19.1%]	21.3%	+3.1%	Fund routing
Procurement Flexibility	17.1% [15.4–18.7%]	30.1%	+0.6%	Least-cost selection
Rarity Multiplier (2×)	16.5% [14.7–18.3%]	34.3%	+2.9%	Price signal
Baseline	16.3% [14.6–17.9%]	34.1%	0% (ref.)	Reference
Price Floor	16.3% [14.7–17.9%]	50.5%	−6.5%	Supply incentive
BCT Precision Mandate	12.7% [11.5–14.3%]	33.0%	−1.3%	Procurement matching
Observed	17.9%	35.4%	n/a	Empirical

References

[1] United Nations Environment Programme. State of finance for nature: The big nature turnaround (2023). URL https://www.unep.org/resources/state-finance-nature-2023.

[2] International Advisory Panel on Biodiversity Credits. Learnings from government-led approaches to nature credit markets (2024). Presented at CBD COP16, October 2024.

[3] zu Ermgassen, S. O. S. E. *et al.* The ecological outcomes of biodiversity offsets under 'no net loss' policies: A global review. *Conservation Letters* **12**, e12664 (2019).

[4] Gibbons, P., Macintosh, A., Constable, A. L. & Hayashi, K. Outcomes from 10 years of biodiversity offsetting. *Global Change Biology* **24**, e643–e654 (2018).

[5] Maron, M. *et al.* Taming a wicked problem: Resolving controversies in biodiversity offsetting. *BioScience* **66**, 489–498 (2016).

[6] Drechsler, M. Flexibility trade-offs in conservation offsets. *Conservation Biology* **38**, e14144 (2024).

[7] zu Ermgassen, S. O. S. E. *et al.* The hidden biodiversity risks of increasing flexibility in biodiversity offset trades. *Biological Conservation* **252**, 108861 (2020).

[8] IPART. Review of the biodiversity offsets scheme: Annual report 2023–24. Tech. Rep., Independent Pricing and Regulatory Tribunal of New South Wales, Sydney, Australia (2024).

[9] Farmer, J. D. Market force, ecology and evolution. *Industrial and Corporate Change* **11**, 895–953 (2002).

[10] Scholl, M. P., Calinescu, A. & Farmer, J. D. How market ecology explains market malfunction. *Proceedings of the National Academy of Sciences* **118**, e2015574118 (2021).

[11] Cont, R. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance* **1**, 223–236 (2001).

[12] IPART. Discussion paper: 2024–25 biodiversity market monitoring review. Tech. Rep., Independent Pricing and Regulatory Tribunal of New South Wales, Sydney, Australia (2025). URL https://www.ipart.nsw.gov.au/sites/default/files/cm9_documents/Discussion-Paper-2024-25-Biodiversity-Market-Monitoring-Review-30-October-2025.pdf.

[13] IPART. Review of the biodiversity offsets scheme: Annual report 2022–23. Tech. Rep., Independent Pricing and Regulatory Tribunal of New South Wales, Sydney, Australia (2023).

[14] Fagiolo, G., Moneta, A. & Windrum, P. A critical guide to empirical validation of agent-based models in economics. *Computational Economics* **30**, 195–226 (2007).

[15] U.S. Army Corps of Engineers & U.S. Environmental Protection Agency. Compensatory mitigation for losses of aquatic resources; final rule (2008).

URL https://www.federalregister.gov/documents/2008/04/10/E8-6918/compensatory-mitigation-for-losses-of-aquatic-resources. 73 Fed. Reg. 19,594–19,705 (April 10, 2008). 33 CFR Parts 325 and 332; 40 CFR Part 230.

[16] Hough, P. & Harrington, R. Ten years of the compensatory mitigation rule: Reflections on progress and opportunities. *Environmental Law Reporter* **49**, 10018–10037 (2019).

[17] Kim, J. E. & Tang, T. Preventing early lock-in with technology-specific policy designs: The Renewable Portfolio Standards and diversity in renewable energy technologies. *Renewable and Sustainable Energy Reviews* **123**, 109738 (2020).

[18] Wood, P. J. & Jotzo, F. Price floors for emissions trading. *Energy Policy* **39**, 1746–1753 (2011).

[19] van Teeffelen, A. J. A. *et al.* Ecological and economic conditions and associated institutional challenges for conservation banking in dynamic landscapes. *Landscape and Urban Planning* **130**, 64–72 (2014).

[20] Bull, J. W., Lloyd, S. P. & Strange, N. Implementation gap between the theory and practice of biodiversity offset multipliers. *Conservation Letters* **10**, 656–669 (2017).

[21] Driscoll, D. A., Banks, S. C., Barton, P. S., Lindenmayer, D. B. & Smith, A. L. Conceptual domain of the matrix in fragmented landscapes. *Trends in Ecology & Evolution* **28**, 605–613 (2013).

[22] Fischer, J. & Lindenmayer, D. B. Landscape modification and habitat fragmentation: a synthesis. *Global Ecology and Biogeography* **16**, 265–280 (2007).

[23] Mace, G. M., Norris, K. & Fitter, A. H. Biodiversity and ecosystem services: a multilayered relationship. *Trends in Ecology & Evolution* **27**, 19–26 (2012).

[24] MacKenzie, D. Is economics performative? option theory and the construction of derivatives markets. *Journal of the History of Economic Thought* **28**, 29–55 (2006).

[25] Novacheck, J. & Johnson, J. X. The environmental and cost implications of solar energy preferences in Renewable Portfolio Standards. *Energy Policy* **86**, 250–261 (2015).

[26] Bush, A., Simpson, K. H. & Hanley, N. Systematic nature positive markets. *Conservation Biology* **38**, e14216 (2024).

[27] Needham, K., de Vries, F. P. & Hanley, N. Designing markets for biodiversity offsets: Lessons from tradable pollution permits. *Journal of Applied Ecology* **56**, 1429–1435 (2019).

[28] Zhou, S., Solomon, B. D. & Brown, M. A. The spillover effect of mandatory renewable portfolio standards. *Proceedings of the National Academy of Sciences* **121**, e2313193121 (2024).

[29] Bjørn, A., Lloyd, S. M., Brander, M. & Matthews, H. D. Renewable energy certificates threaten the integrity of corporate science-based targets. *Nature Climate Change* **12**, 539–546 (2022).

[30] Donna, J. D. & Espín-Sánchez, J.-A. The illiquidity of water markets. *The Review of Economic Studies* rdag004 (2026).

[31] Connor, J. D., Franklin, B., Loch, A., Kirby, M. & Wheeler, S. A. Trading water to improve environmental flow outcomes. *Water Resources Research* **49**, 4265–4276 (2013).

[32] Lo, A. W. The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *The Journal of Portfolio Management* **30**, 15–29 (2004).

[33] Haldane, A. G. & May, R. M. Systemic risk in banking ecosystems. *Nature* **469**, 351–355 (2011).

[34] May, R. M., Levin, S. A. & Sugihara, G. Complex systems: Ecology for bankers. *Nature* **451**, 893–895 (2008).

[35] Volterra, V. Variazioni e fluttuazioni del numero d'individui in specie animali conviventi. *Nature* (1926). Summary.

[36] Hardin, G. The competitive exclusion principle. *Science* **131**, 1292–1297 (1960).

[37] Pratt, J. W. Risk aversion in the small and in the large. *Econometrica* **32**, 122–136 (1964).

[38] Brock, W. A. & Hommes, C. H. A rational route to randomness. *Econometrica* **65**, 1059–1095 (1997).

[39] Chiarella, C. & He, X.-Z. Heterogeneous beliefs, risk and learning in a simple asset pricing model. *Computational Economics* **19**, 95–132 (2002).

[40] Arrow, K. J. & Lind, R. C. Uncertainty and the evaluation of public investment decisions. *American Economic Review* **60**, 364–378 (1970).

[41] Drechsler, M. On the cost-effective temporal allocation of credits in conservation offsets. *Environmental and Resource Economics* **82**, 437–459 (2022).

[42] Audit Office of New South Wales. Effectiveness of the biodiversity offsets scheme (2022). URL https://www.audit.nsw.gov.au/our-work/reports/effectiveness-of-the-biodiversity-offsets-scheme.

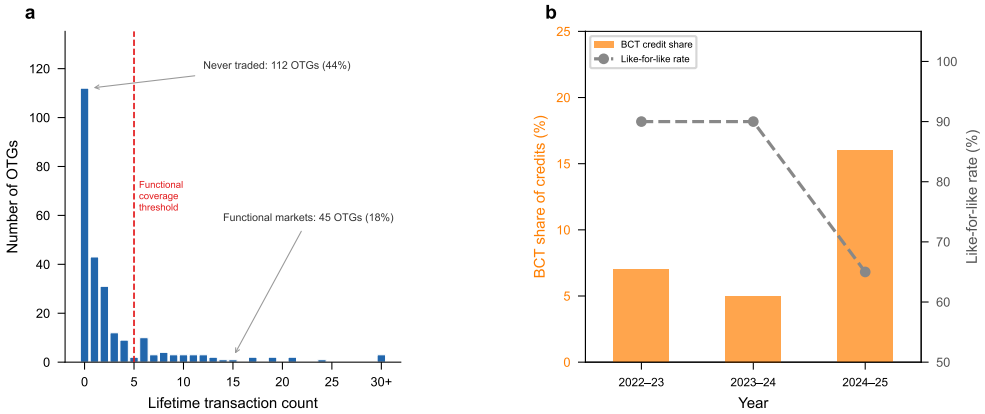

Fig. 1 The coverage gap in the NSW biodiversity offset market. a, Distribution of 252 ecosystem OTGs by lifetime transaction count; 112 (44%) have never traded; only 45 (18%) exceed the functional coverage threshold of 5 transactions (red dashed line). **b**, BCT market share (orange bars) and like-for-like purchase rate (grey dashed line); share increased from 7% to 16% while like-for-like precision fell from 90% to 65%.

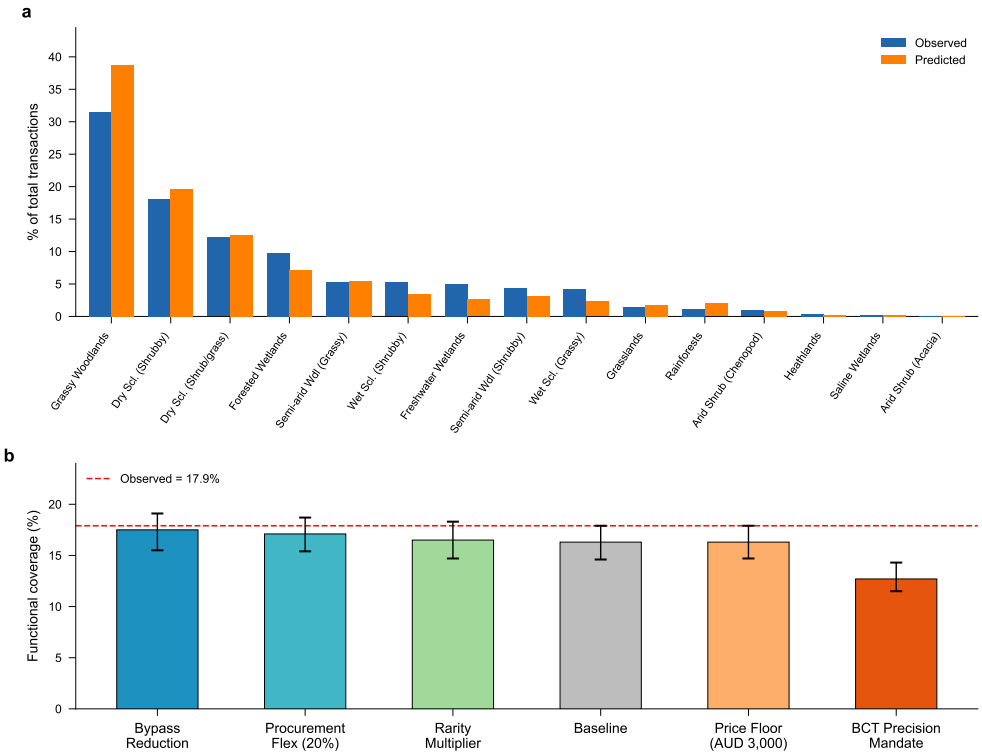

Fig. 2 Formation-level model: validation and policy outcomes. a, Observed vs predicted transaction distribution across 15 vegetation formations (Spearman $\rho = 0.98$); the model reproduces the concentration of demand in Grassy Woodlands and Dry Sclerophyll Forests without targeting these formations during calibration. **b**, Functional coverage under the six policy scenarios of Table 1 (ABM, 50 Monte Carlo seeds; error bars: 10th–90th percentile; red dashed line: observed = 17.9%). Bypass reduction and procurement flexibility raise coverage by keeping demand in the open market; price floors and the BCT precision mandate reduce it.

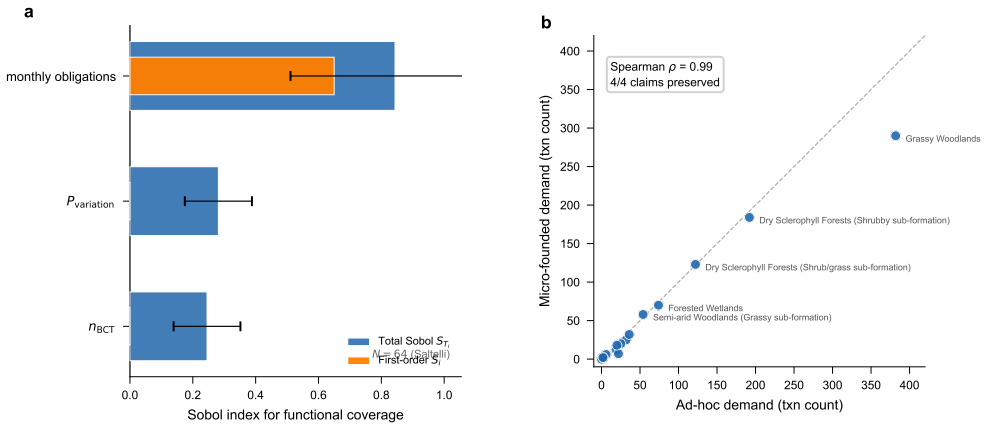

Fig. 3 Robustness: global sensitivity analysis and demand-function specification. a, Sobol indices for functional coverage as the quantity of interest. Total Sobol index S_{T_i} (blue bars) measures the share of output variance attributable to parameter i including all interactions; first-order S_i (orange bars) measures the share attributable to parameter i alone. Whiskers: bootstrap 95% confidence intervals ($B = 1{,}000$). Saltelli design with N base samples per parameter; full design and indices in Supplementary Note 2 and Supplementary Table 9. **b**, Demand-function specification test: per-formation transaction counts produced by the ad-hoc bid rules (horizontal axis) and the CARA-derived implementation (vertical axis), Spearman $\rho \geq 0.97$ across 50 Monte Carlo seeds. The two specifications produce equivalent outputs at the formation level.